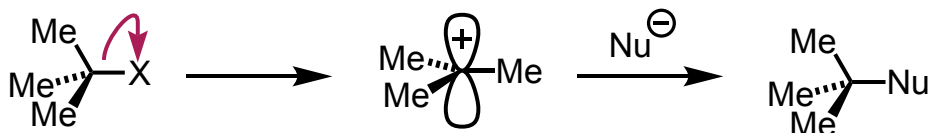
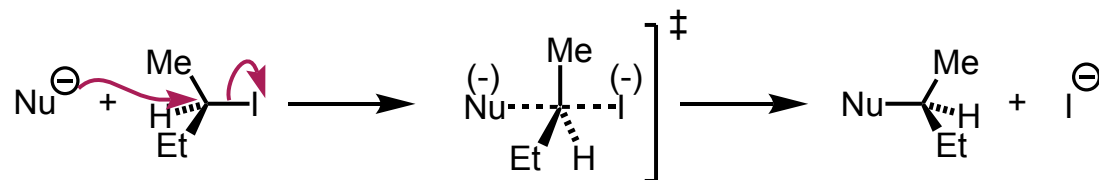


The nature of pericyclic reactions

- Ionic reactions – may or may not have an intermediate



- $\text{S}_{\text{N}}1$ reactions are stepwise and have a defined intermediate



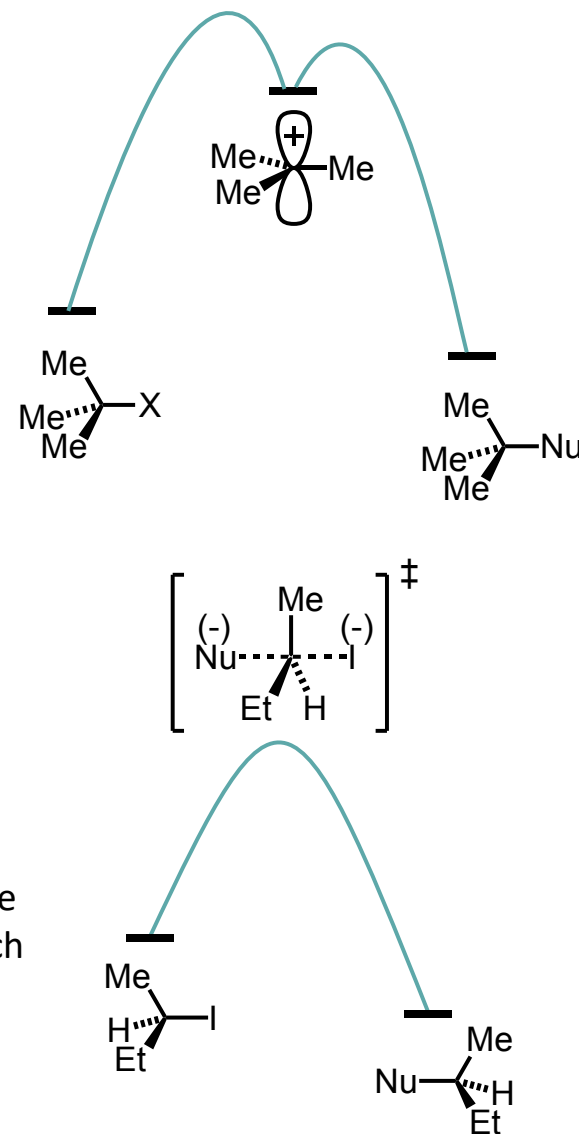
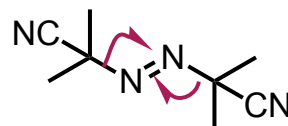
- $\text{S}_{\text{N}}2$ reactions are *concerted* and have no intermediate

■ In ionic reactions the curly arrows identify where the electrons have come from and where they are going and which bonds have been made and which broken.

- Radical reactions involve the correlated movement of single electrons



frequently shortened to



Pericyclic Reactions

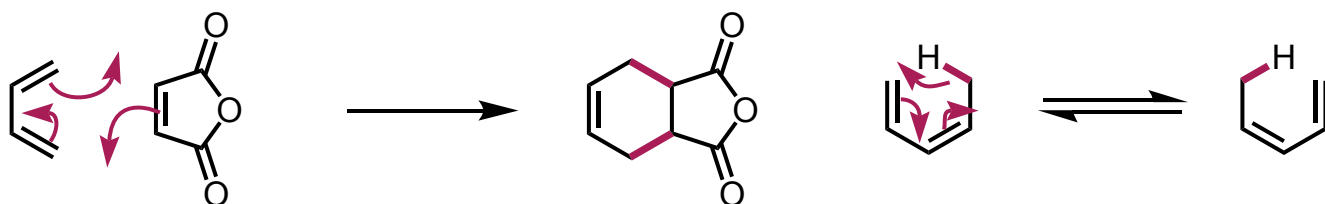
- Third distinct class of reactions

Pericyclic reactions – Reactions in which all first-order changes in bonding relationships take place in concert on a closed curve. *R. B. Woodward and R. Hoffmann 1969.*

i.e. the reactions have cyclic transition states in which all bond-forming and bond-breaking take place in a concerted manner without the formation of an intermediate.

- Pericyclic reactions involve a transition state with a cyclic array of interacting orbitals; a reorganisation of σ and π -bonds occurs within this cyclic array.

- Originally termed 'no mechanism reactions' – they could not be explained by standard nucleophile/electrophile mechanisms.



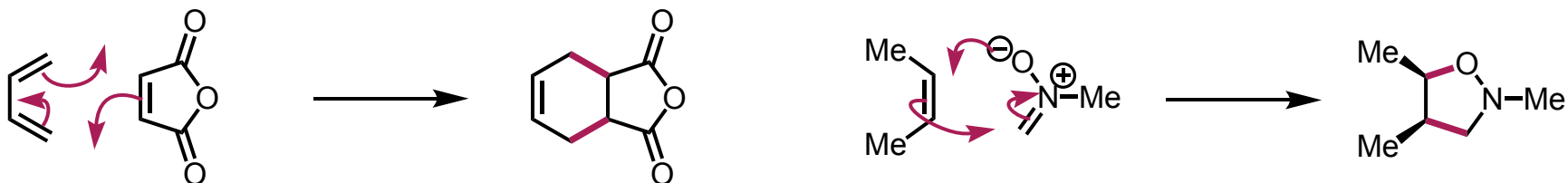
- No absolute sense in which the electrons flow from one component to another; however, sometimes it is more sensible to push the arrows in only one direction.

- Here curly arrows are used to show which bonds are being made/broken rather than the direction of flow of electrons.

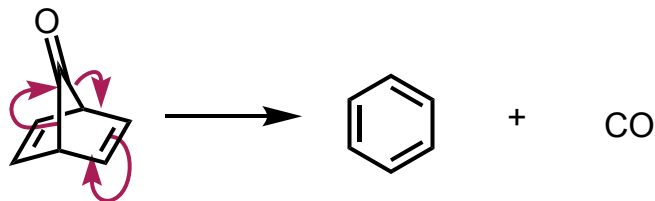


Pericyclic Reactions – **four classes**.

■ Cycloadditions - **Two components** come together to form two new σ -bonds at the ends of both components and joining them together to form a **ring**.



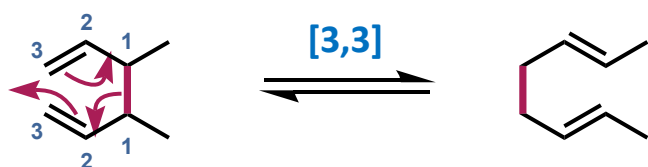
■ Cheletropic reactions are a sub-class of cycloaddition reactions in which the **two σ -bonds** are made or broken to the **same atom**.



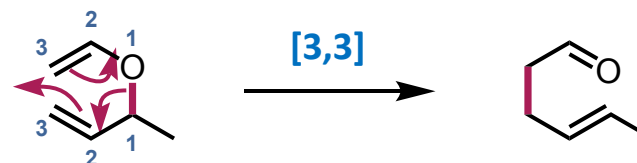
■ Electrocyclic Reactions – **Unimolecular** reactions characterised by the **formation of a ring** from an open chain conjugated system with a σ -bond forming across the ends of the conjugated system.



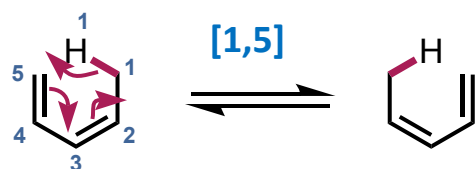
■ **Sigmatropic rearrangements** – **Unimolecular isomerisations** which formally involve the overall movement of a σ -bond from one position to another.



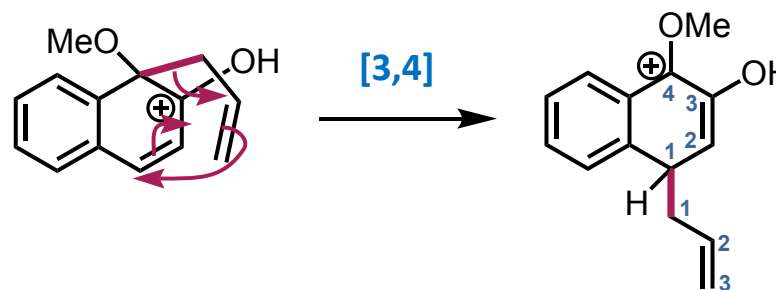
Cope rearrangement



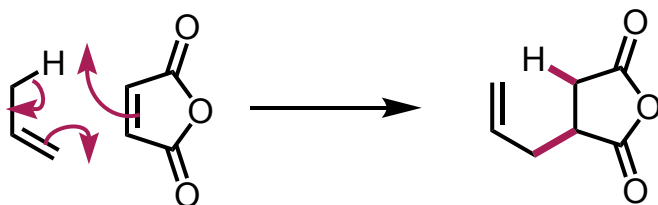
Claisen rearrangement



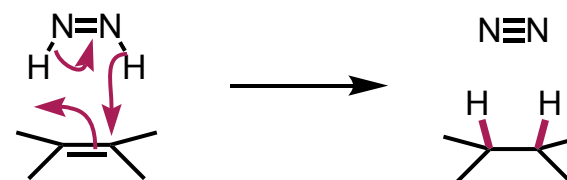
[1,5]-hydride shift



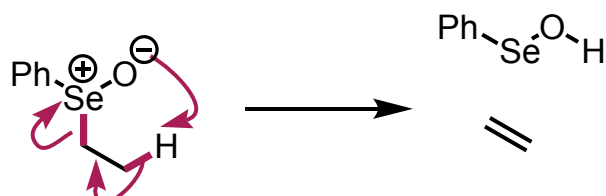
■ **Group transfer** – appear to be a mix of a sigmatropic rearrangement and a cycloaddition. They are **bimolecular** and so are not sigmatropic rearrangements, and **no ring is formed** so they are not cycloadditions.



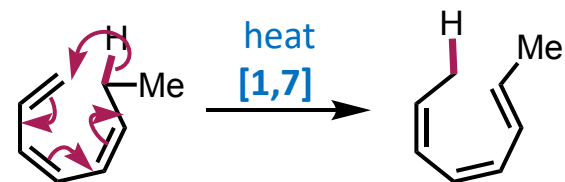
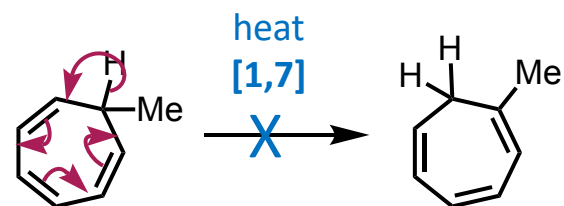
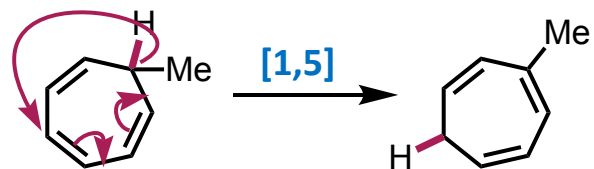
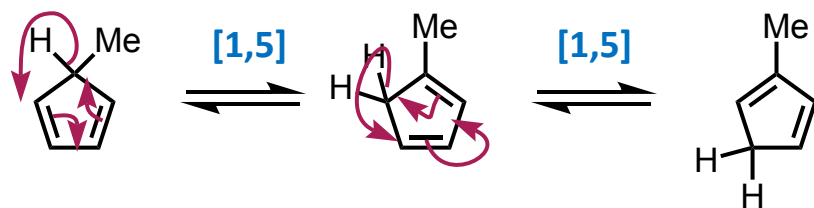
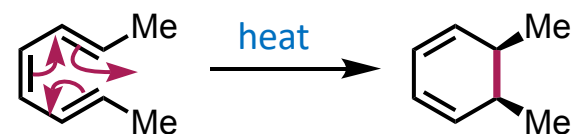
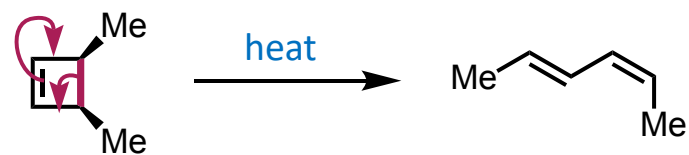
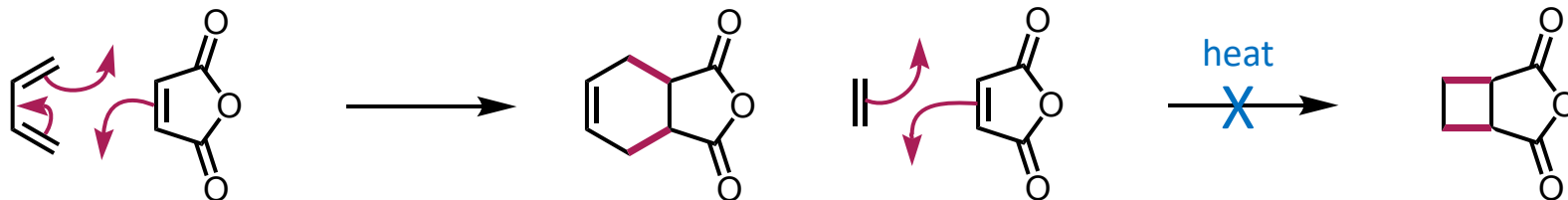
Alder-ene - reactions



diimide reduction



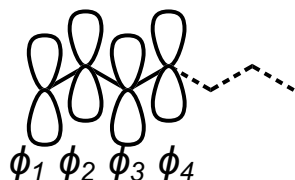
■ Require a theory/theories to explain these results



- Hückel Molecular Orbital Theory for Linear π -systems.
- Applicable to conjugated planar cyclic and acyclic systems.
- Only the π -system is included; the σ -framework is ignored (in reality σ -framework affects π -system).
- Used to calculate the wave functions (ψ_k) and hence *relative* energies by the LCAO method.

$$\text{MO} = \psi_k, \text{AO} = \phi_i \text{ and } \psi = \sum_{i=1}^n c_{ki} \phi_i \quad \text{i.e. } \psi_k = c_1 \phi_1 + c_2 \phi_2 + c_3 \phi_3 + c_4 \phi_4 + c_5 \phi_5 \dots$$

k = MO number ($1 \rightarrow n$); i = atom position in π system ($1 \rightarrow n$); n = number of p-orbitals



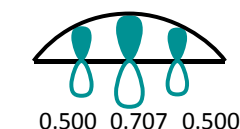
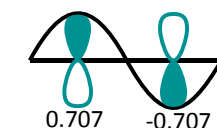
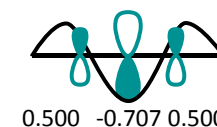
For n contributing p-orbitals there will be n molecular orbitals
 $c_{ki} = (2/(n+1))^{1/2} \cdot \sin(\pi k i / (n+1))$

- For each MO ψ_k $\sum c_i^2 = 1$ (normalisation) i.e. the sum of the squares of the coefficients for any MO must be 1.

- For each AO ϕ_i $\sum c_k^2 = 1$ i.e. the sum of the squares of the coefficients for any AO over all MOs must be 1.

- Each MO must be symmetric or antisymmetric with respect to any symmetry operation of the molecule.

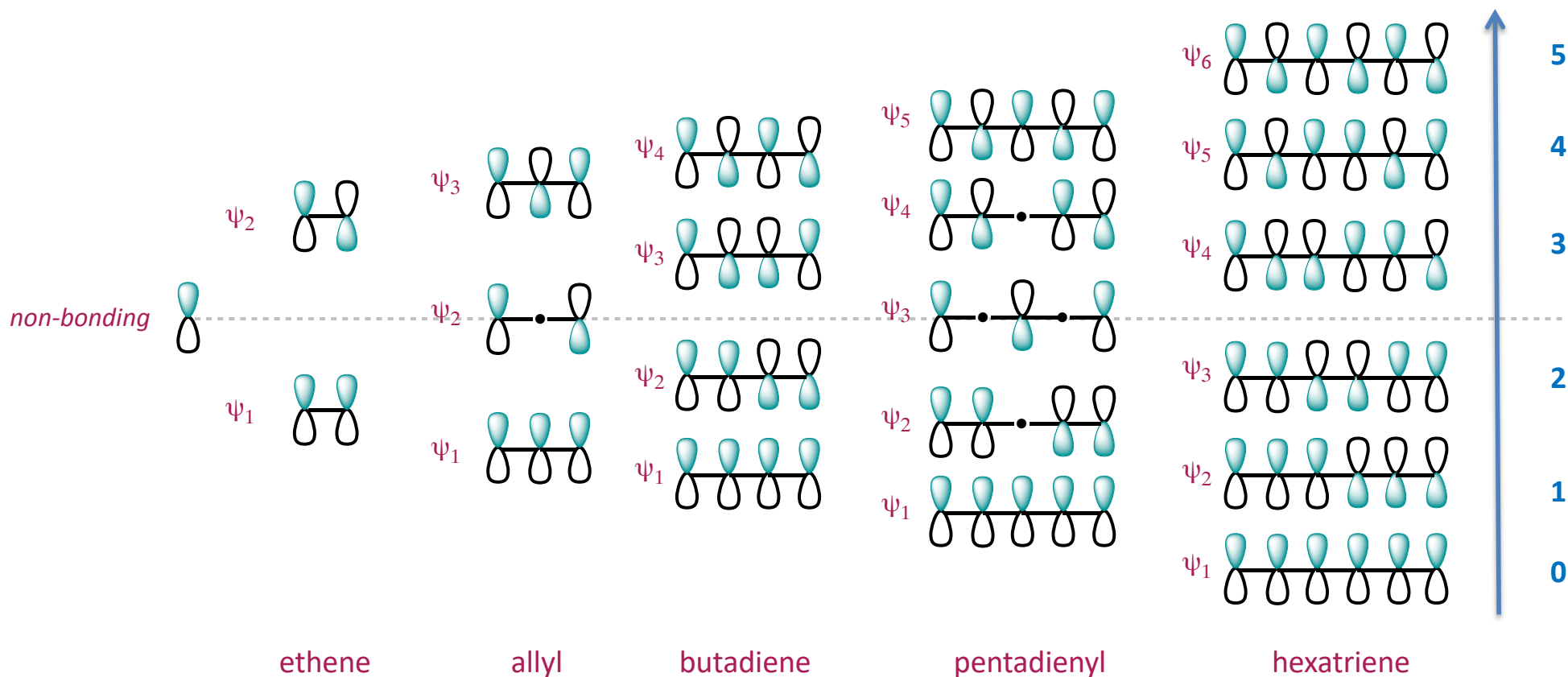
- Molecular Orbital Energies = $E_k = \alpha + m_j \beta$ $m_j = 2 \cos[j\pi/(n+1)]$ $j = 1, 2, \dots, n$



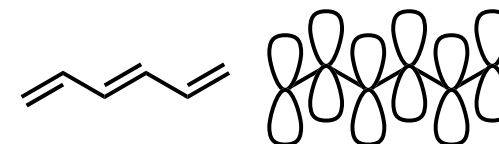
■ To draw a molecular orbital diagram:

- i) count the number of p-orbitals – n
- ii) count the number of electrons – π -bond = 2, unpaired electron = 1, carbanion = 2, carbocation = 0
- iii) draw n horizontal lines stacked on top of one another to represent the MOs
- iv) draw the molecular orbitals as the combination of p-orbitals with an increasing number of nodes from 0 for ψ_1 to $n-1$ for ψ_n such that **each MO is symmetric or antisymmetric with respect to any symmetry operation of the molecule.**
- v) if the number of nodes is even then the terminal orbital coefficients will be the same phase
- vi) if the number of nodes is odd then the terminal coefficients will be of opposite phase

increasing
number of nodes



- 1,3,5-Hexatriene
- six 2p atomic orbitals give 6π molecular orbitals; $n = 6, j = 1, 2, 3, 4, 5, 6$



$$m_j = 2\cos(\pi/7) = 1.80$$

$$2\cos(2\pi/7) = 1.25$$

$$2\cos(3\pi/7) = 0.45\ldots$$

and the corresponding negative values

$$\text{energy } E = \alpha + 1.80\beta$$

$$\alpha + 1.25\beta$$

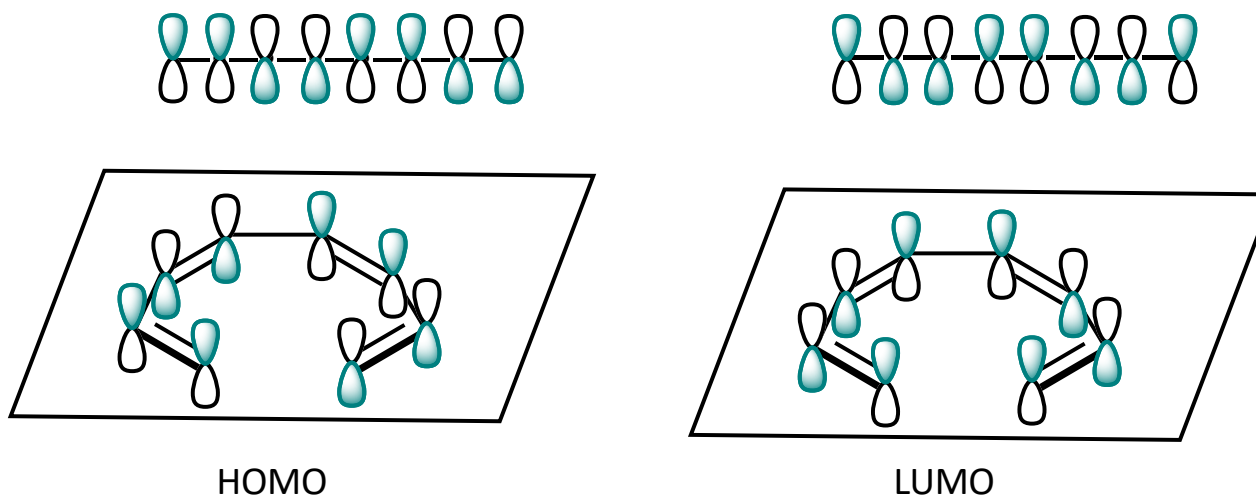
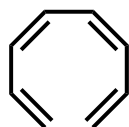
$$\alpha + 0.45\beta$$

$$\alpha - 0.45\beta\ldots\text{etc}$$

MO		no. nodes	energy		HMOT MO (calculated)	
ψ_6		5	$\alpha - 1.80\beta$	—		
ψ_5		4	$\alpha - 1.25\beta$	—		
ψ_4		3	$\alpha - 0.45\beta$	—		
ψ_3		2	$\alpha + 0.45\beta$	$\updownarrow$		
ψ_2		1	$\alpha + 1.25\beta$	$\updownarrow$		
ψ_1		0	$\alpha + 1.80\beta$	$\updownarrow$		

■ stabilisation energy = $E_{\text{stab}} = 2(3\alpha + 3.5\beta) = 6\alpha + 7\beta$ remember $E_{\text{stab}} (\text{ethene}) = 2\alpha + 2\beta$

- The HOMO of (neutral) polyenes can be readily constructed by putting nodes on the single bonds; hence the HOMO appears to be alternating isolated π -bonding pairs.
- The LUMO of (neutral) polyenes can be constructed by beginning with an 'isolated' p-orbital and then alternating 'isolated' π -bonding pairs and ending with an 'isolated' p-orbital.
- The HOMO and LUMO are termed the Frontier Orbitals.



Correlation Diagrams

- During a pericyclic reaction the orbitals of the starting material are smoothly converted into the orbitals of the product.
- This means that the symmetry of the orbitals with respect to any symmetry operations of the molecule must be conserved in moving from the starting material(s) to product – this is the 'Conservation of Orbital Symmetry', which is readily depicted in an 'orbital correlation diagram'